

MID 3040 Final Exam Task

Goal:

Develop a tool to minimize the Life Cycle Assessment (LCA) impact of a wall assembly based on a given 3D model. The optimization should be performed using a database of possible materials for each layer of the wall assembly.

Constraints:

- Target U-value: $0.14 \text{ W}/(\text{m}^2\text{K}) \pm 10\%$

Phase 1 – Material Database Preparation:

- Research 10 Environmental Product Declarations (EPDs) relevant to your assigned wall layer.
- Fill in the provided JSON template to create the material database for use in the optimization process.

Phase 2 – Tool Development:

- Develop the optimization algorithm/tool that selects the best material combination to minimize LCA while meeting the thermal constraint.

Deliverables:

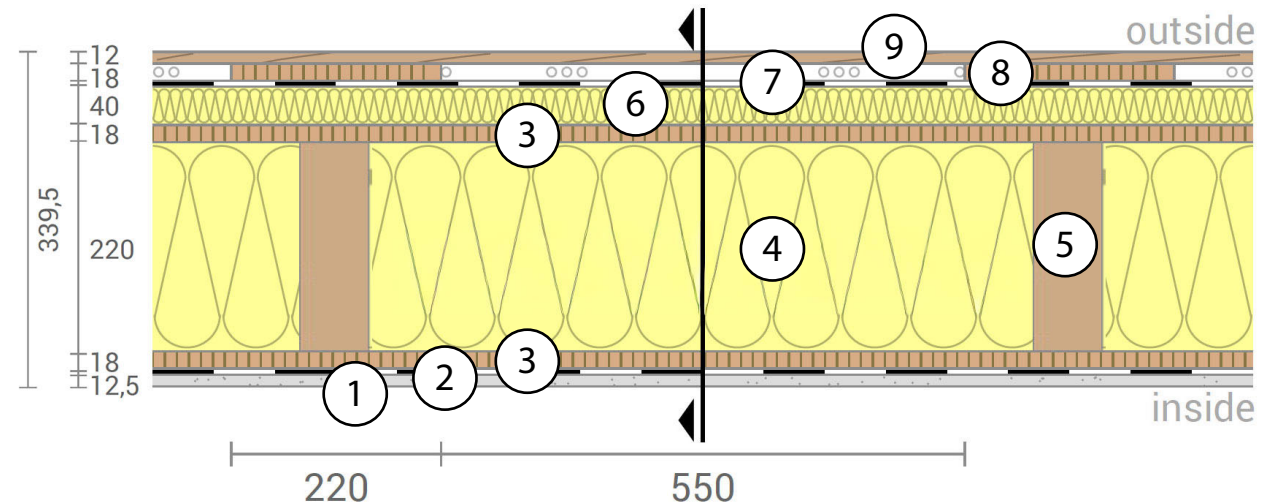
- Optimization Tool / Algorithm
- Technical Documentation
 - Include a theoretical explanation of the algorithm used
 - Describe the architecture and workflow of your tool
 - Format: A4, portrait

Presentation

- Summarize your concept, results, and process
- Format: PowerPoint or PDF, A3 landscape

README File

- Include installation or setup instructions, if needed



- 1 Internal Finishing Gypsum board
- 2 Vapor control layer
- 3 Wood-based board
- 4 Cavity Insulation
- 5 Primary Structure
- 6 Second insulation layer
- 7 Weather control layer
- 8 Battens - rainscreen
- 9 Facade cladding